ScholarLens :Transforming Scientific Content into Accessible Multimedia Project Report

CSAI 810: Topics in Artificial Intelligence

Under Supervision of:

Dr. Rahatara Ferdousi

By:

Aya Mohamed Mahmoud

Fatma Ahmed Mansour

MercuryLens: Predictive Modeling of Chemical Contaminants in Ontario Fish

Group_36

Abstract

This project presents **MercuryLens**, a machine learning framework designed to predict fish consumption advisory levels (Safe, Moderate, and High Risk) for Ontario fishes. Utilizing the 2025 Ontario Fish Advisory Database (108,000 records)[1], we transitioned from traditional descriptive lookup tables at Current advisory systems (Ontario.ca)[2] to an automated predictive system. Our **Stacking Ensemble** delivered performance with a final accuracy of **83.25%** and a **0.83 F1-score** for high-risk detection, providing a robust tool for public and food health in Ontario.

Introduction

Fishing is a cornerstone of the Canadian economy, with a projected government investment of **$1.41 billion** for 2025-2026[3]. However, public health is increasingly threatened by bio-accumulative toxins like mercury and PCBs in Ontario's 250,000 lakes.

MercuryLens addresses this gap by transforming spatial (Latitude/Longitude) and biological (Species, Length) data into predictive insights. Inspired by a marine studies (Piccialli et al., 2022)[4], we aim to provide automated consumption guidelines, protecting vulnerable groups "children and pregnant women".

Objective

The core objective of **MercuryLens** is to develop a machine learning system that predicts fish safety. This is achieved by applying **Geospatial Scaling** to identify "Pollution Pockets," evaluating **tree based** models, and prioritizing **Recall for high-risk categories** to ensure maximum public health safety for Ontario's residents.

Method

To ensure replicability, our methodology followed a structured pipeline:

1. **Preprocessing and Cleaning**
   Irrelevant Feature Removal: We eliminated features with zero variance (e.g., Guide Year) and redundant identifiers (e.g., Waterbody Group, Species Code) to reduce model complexity.
   Statistical Imputation of Missing Values: We identified 7,962 missing records within the 'Contamination Cause' column. It was observed that these records exclusively mapped to (Level 32). Consequently, these were logically imputed as 'Safe'.
   Geospatial Scaling: We applied a scaling factor (divided by **10,000**) to restore them to standard geographic coordinates (e.g., 42° to 56° N).

2. **Feature Engineering**
   Numerical Transformation: We used regular expressions to convert text-based length ranges (e.g., '40-45 cm') into a continuous numerical feature (Mean_Length), enabling the model to treat fish size as a quantitative risk driver.
   Spatial Intelligence: We implemented Coordinate Binning by rounding GPS data to the nearest 0.1 degree to create localized grid_id clusters.

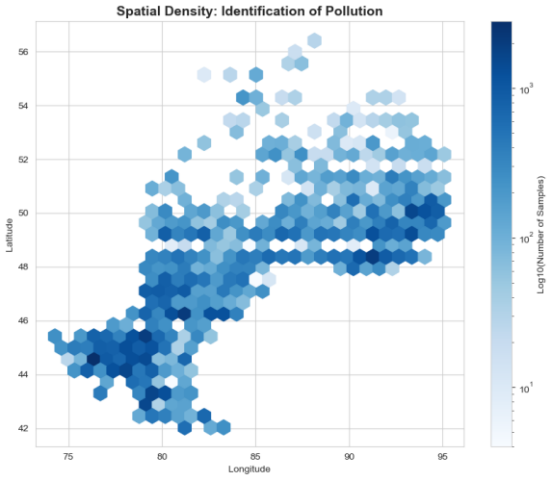

Image 1: Spatial Density: Identification of Pollution

3. **Dual-Branch Pipeline Strategy**
   Encoded Branch: Implemented **Label Encoding** for the **Random Forest Baseline** and **XGBoost**.
   Native Branch: Preserved raw categorical strings for **CatBoost** to leverage its native symmetric tree processing, which is highly efficient for high-cardinality spatial features.

Experiments

We conducted four iterative experiments using the Anaconda platform and Jupyter Notebook to evolve our system from a simple baseline to an optimized ensemble:

1. **Baseline Benchmarking:** We first trained a Random Forest (RF) model to establish a performance floor. While RF provided a stable baseline of 76% accuracy, error analysis revealed a significant struggle with the 'Moderate Risk' class (Recall: 0.59), often misclassifying it as 'Safe'.
2. **Boosting Showdown:** We compared CatBoost and LightGBM to evaluate their handling of our engineered spatial features. LightGBM demonstrated superior sensitivity to the complex non-linear patterns in the geospatial data, pushing accuracy beyond the 80% threshold.
3. **Hyperparameter Optimization:** To maximize performance, we performed Grid Search on our top models. We specifically tuned num_leaves and max_depth to balance model complexity with generalization, successfully reducing overfitting in the High-Risk category.
4. **Error Analysis & Stacking:** Utilizing Confusion Matrices, we identified that most errors occurred at the decision boundaries between 'Moderate' and 'High' risk. To mitigate this, we implemented a Stacking Ensemble, which fused the diverse strengths of our models to stabilize predictions.

5. **Key Outcome:** The transition from traditional bagging (RF) to advanced gradient boosting significantly improved the recall of toxic categories, which is a critical requirement for a public health safety tool.

6. **Evaluated Techniques**

   We benchmarked a diverse set of architectures to ensure robust predictions:

   - **Random Forest (RF):** Established as our bagging-based **baseline**.
   - **Gradient Boosting Ensemble (XGBoost & LightGBM):** Evaluated for high-speed performance; **LightGBM** was specifically selected for its **leaf-wise growth strategy**, which effectively captures complex environmental patterns.
   - **CatBoost:** Utilized for its superior **native handling** of high-cardinality categorical features.
   - **Stacking Ensemble:** A final **Meta-Model (Logistic Regression)** was implemented to fuse the predictive strengths of the top-performing models.

Results

The following results directly relate to our goal of ensuring food safety:

Performance Comparison	precision	Recall	F1-Score	Accuracy
Random Forest (Baseline)	0.76	0.76	0.76	76.00%
XGBoost	0.73	0.74	0.74	74.00%
CatBoost	0.83	0.81	0.82	81.24%
LightGBM	0.83	0.83	0.83	83.14%
Extra Trees	0.73	0.71	0.72	71.37%
Stacking Classifiers	0.83	0.83	0.83	83.25%

Table 1: Performance Comparison

Critical Detection: The stacking classifiers model achieved a **0.90 Precision** for High-Risk categories**.**

Models Comparison

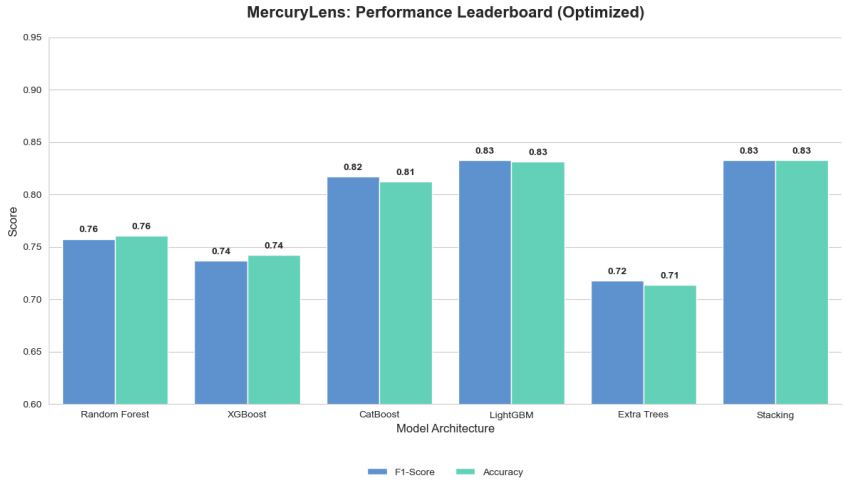

Image 2: Models Comparison.

Confusion Matrix

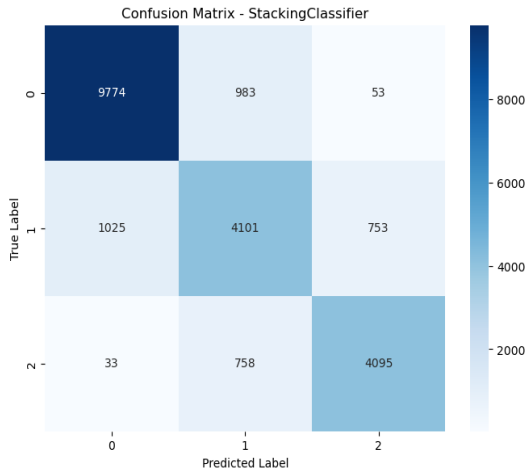

Image 3: Confusion Matrix.

Project Timeline

1. **Week 1:** Data Acquisition and rigorous cleaning of the Ontario Fish Advisory Database.
2. **Week 2-3:** Feature Engineering, specifically focusing on Geospatial Scaling and Spatial Analysis.
3. **Week 4:** Model Iteration, comparing various architectures, and conducting Hyperparameter Tuning.
4. **Week 5:** Final Evaluation, error analysis using confusion matrices, and report documentation.

Conclusion

The hypothesis that machine learning can automate environmental advisories was validated. **MercuryLens** significantly outperforms traditional descriptive methods by providing an **83% accurate** safety prediction.

- **Shortcomings:** The model still shows some confusion in "Moderate" risk levels due to the inherent overlap in chemical data.
- **Implications:** This system can be integrated into mobile apps for real-time safety checks.
- **Future Work:** Integrating real-time water temperature and industrial proximity data could further push accuracy toward 90%.

References

[1]*https://data.ontario.ca/dataset/guide-to-eating-ontario-fish-advisory-database/resource/bbc157b9-309c-4d5c-9403-a6f71d503451*
[2] *https://www.ontario.ca/page/guide-eating-ontario-fish#section-3*
[3] *2025-26 Departmental Plan*
[4] *https://ieeexplore.ieee.org/document/10031227*